BJT Amplifier

ENGS152 Circuits Lab

27 March 2026

Step 1: Circuit Design

1. $V_{cc} = 10\ V\ (Nahapetyan)$
2. $\beta = 100$
3. $I_C = 2\ mA$
4. $V_c = V_{cc}/2$
5. $V_E = V_{cc}/10$
6. $I_E = I_c$
7. $I_{BIAS} > 10 * I_B$
8. $V_{BE} = 0.7\ V$
9. $V_T = 25\ mV$
10. **2N3904** transistors to be used

$$R_L = \frac{V_{cc} - V_c}{I_c} = \frac{10 - 5}{0.002} = \boldsymbol{2.5\ k\Omega}$$

$$R_E = \frac{V_E}{I_E} = \frac{1}{0.002} = \boldsymbol{500\ \Omega}$$

$$V_B = V_E + V_{BE} = 1 + 0.7 = \boldsymbol{1.7\ V}$$

$$I_B = \frac{I_C}{\beta} = \frac{0.002}{100} = \boldsymbol{20\ \mu A}$$

$$I_{BIAS} = 10 \times I_B = \boldsymbol{200\ \mu A}$$

$$I_{BIAS} = 200\ \mu A > I_B$$

$$R_2 = \frac{V_B}{I_{BIAS}} = \frac{1.7}{200 \cdot 10^{-6}} = \boldsymbol{8.5\ k\Omega}$$

$$R_1 = \frac{V_{CC} - V_B}{I_{BIAS}} = \frac{10 - 1.7}{200 \cdot 10^{-6}} = \boldsymbol{41.5\ k\Omega}$$

$$r_e = \frac{V_T}{I_E} = \frac{25 \times 10^{-3}}{2 \times 10^{-3}} = \boldsymbol{12.5\ \Omega}$$

$$A_v = -\frac{R_c}{r_e} = \frac{2.5 \times 10^{-3}}{12.5} = \mathbf{-200}$$

$$\beta r_e = 100 \times 12.5 = \mathbf{1250\ \Omega}$$

$$R_{in} = (\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{\beta r_e})^{-1} = \left(\frac{1}{41.5 \times 10^3} + \frac{1}{8.5 \times 10^3} + \frac{1}{1250}\right)^{-1} \approx \mathbf{1061.86}$$

$$\boldsymbol{f_c = 20\ H_z}$$

$$C_1 = \frac{1}{2\pi R_{in} f_c} = \frac{1}{2\pi \cdot 1062 \cdot 20} \approx \mathbf{7.5\ \mu F}$$

$$X_C \leq \frac{R_E}{10}$$

$$X_C = \frac{1}{2\pi f C}$$

$$\frac{1}{2\pi f C_2} \leq \frac{R_E}{10}$$

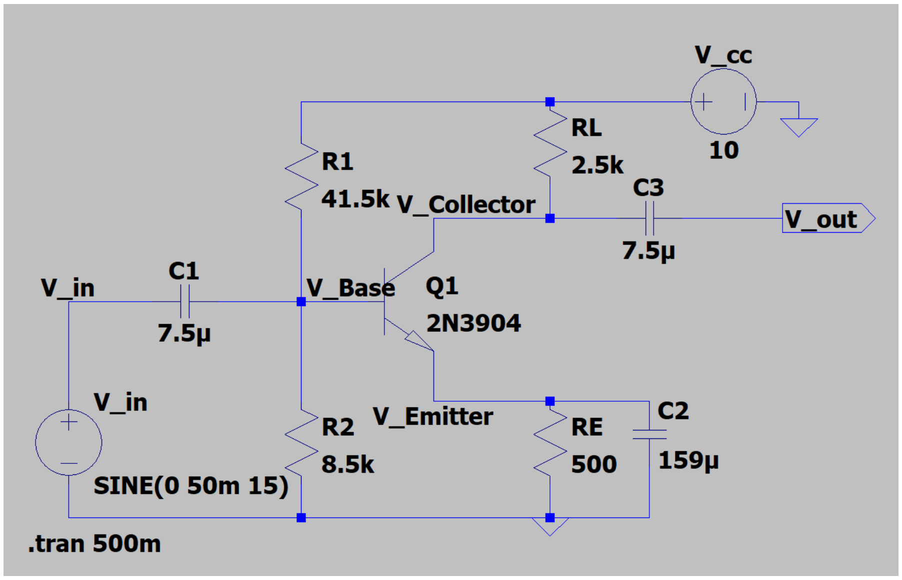
$$C_2 \geq \frac{10}{2\pi f R_E} = \frac{10}{2\pi * 20 * 500} \approx \mathbf{159\ \mu F}$$

$$C_3 = C_1 = \mathbf{7.5 \mu F}$$

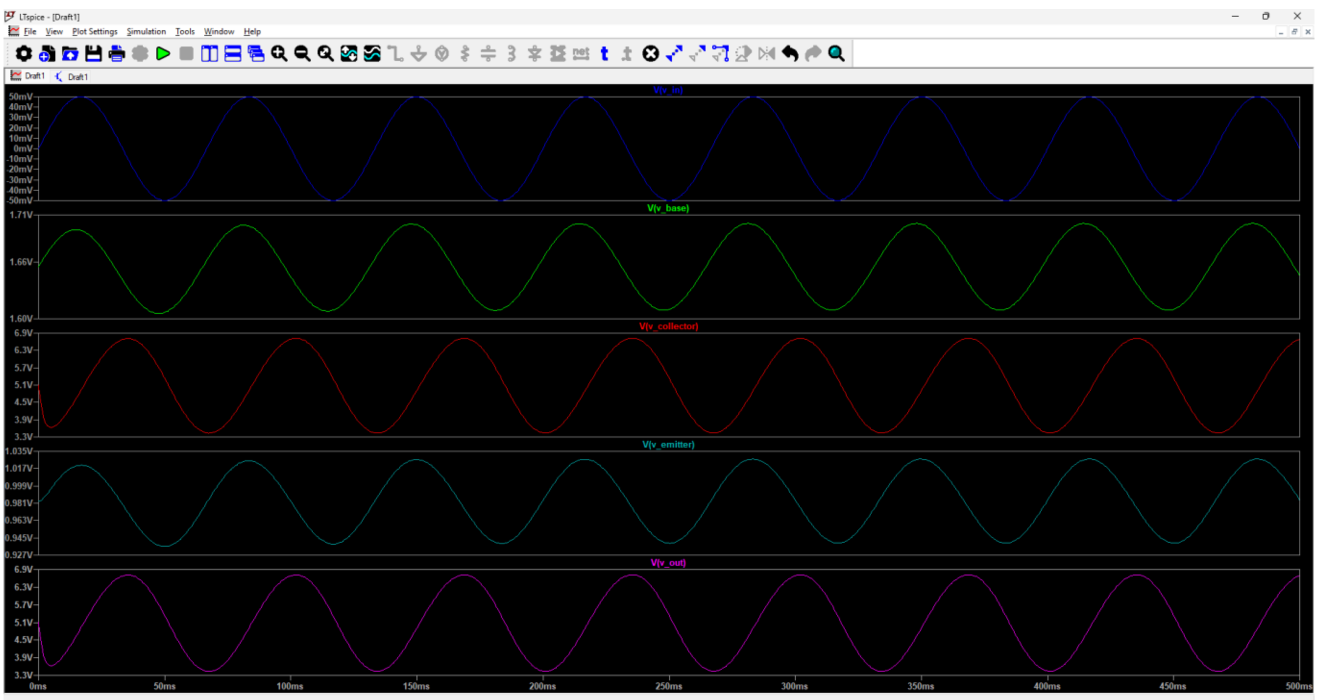
Step 2: Circuit Simulation

Now we are asked to simulate the circuit based on the calculations from Step 1. Input must be a sine wave with an amplitude of 50 mV. A coupling capacitor on the output was added to remove the DC offset on the collector. We have to test the amplifier's performance across three different frequency ranges relative to the calculated cutoff frequency.

Below you can see the circuit configuration after inserting the components based on the calculations from step 1.

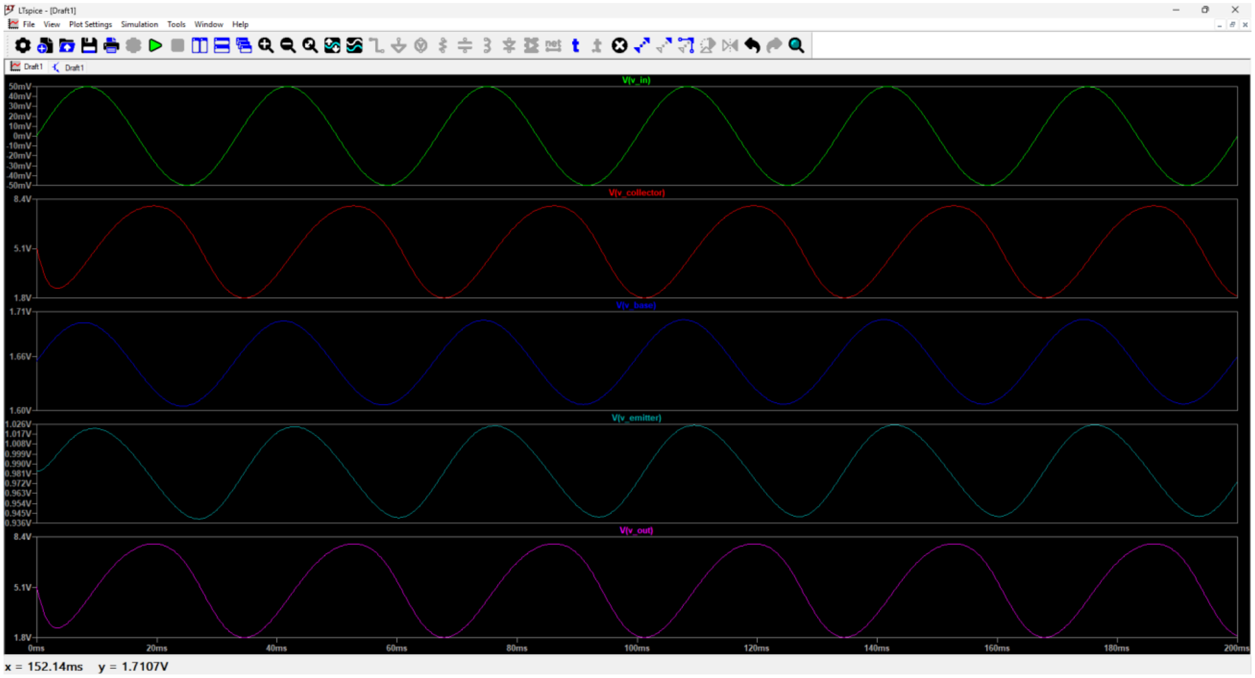


Below Cutoff Frequency (f < fc) 15Hz:

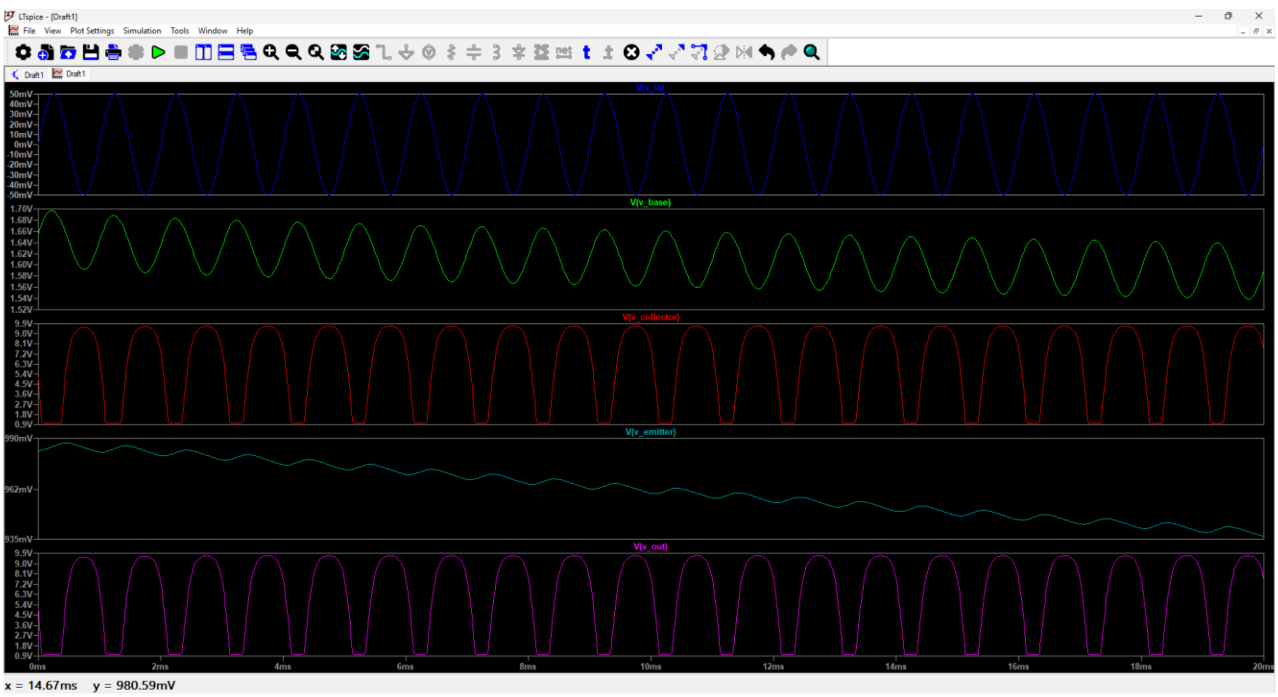


	Theory	Observed
V_{Base}	1.7 V	≈ 1.66 V
$V_{Collector}$	5 V	≈ 5.1 V
$V_{Emitter}$	1 V	≈ 0.981 V

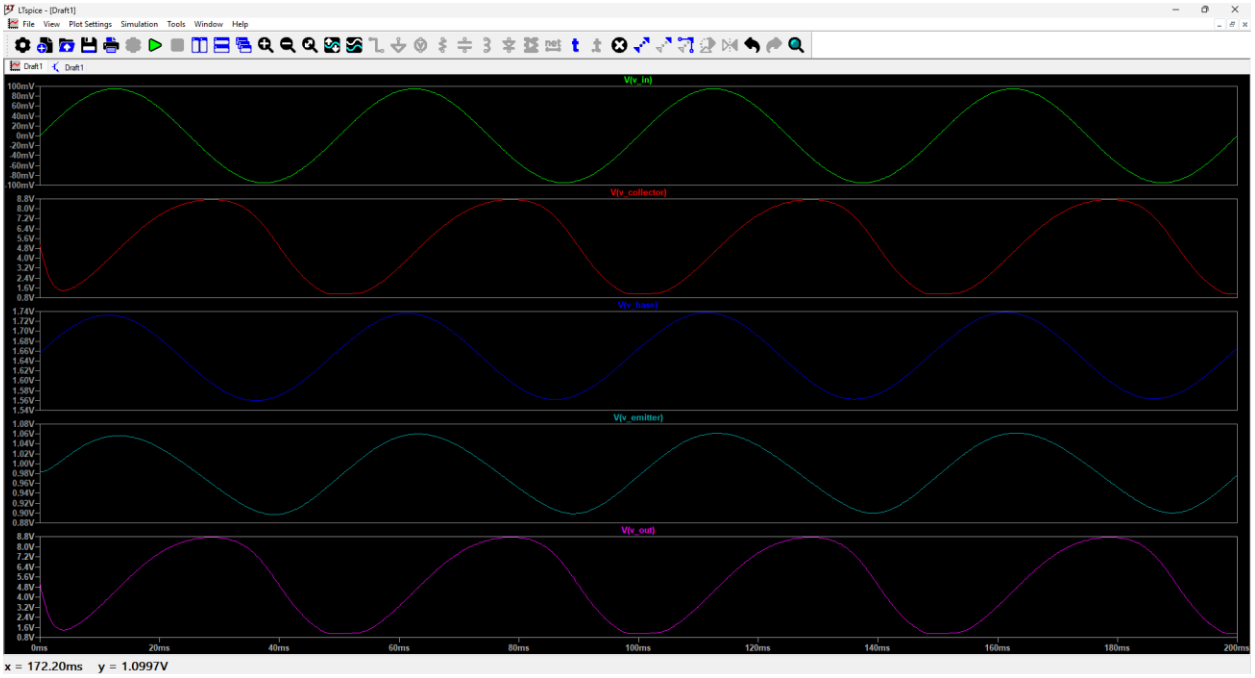
Slightly larger than the cutoff frequency (f ≥ fc) 30Hz:



Much larger than the cutoff frequency (f >> fc) 1 kHz:



Notice that the input and output signals have a phase shift of 180 degrees, meaning the signal has got inverted, also the output voltage has significantly larger amplitude. As we can see from the plots, when the frequency is around the cutoff frequency, the output beings to increase, but is attenuated. However, when the frequency is many times larger than the cutoff frequency, the signal becomes amplified.



We can see the output voltage starting to clip around frequency of 90 mV voltage. The clipping happens when an amplifier or system is pushed beyond its maximum capacity, causing the peaks of signal waveform to be flattened, which we can notice in the plot. Vc is selected to be 50% of Vcc to make it the midpoint and ensure signal has room from both upper and lower limits, this prevents from clipping on one side.